# CSE 708
# Graph Theory

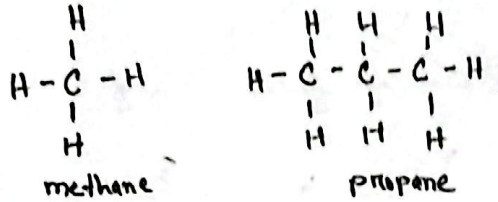
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$X = \text{mod}(21301002, 7) + 1$

$= 3$

1.3

Answer: (i)




From the chemical structure we can make the following observation:

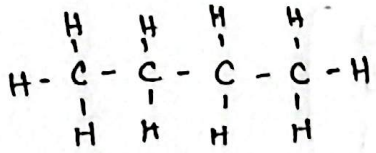
Carbon Vertices (C): Every carbon vertices has a degree of 4. $\deg(C) = 4$. This corresponds to carbons chemical property of having 4 electrons for bonding.

Hydrogen Vertices (H): Every hydrogen vertices has a degree of 1. $\deg(H) = 1$. They are called leaf nodes in graph theory.

(ii) There are two different chemical molicules with formula $C_4H_{10}$.

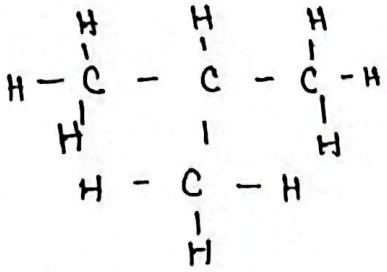
Drawing the graph corresponding to the structure -

1.



n-butane (Straight Chain Isomer)

2.



2-methylpropane (Branched Chain Isomer)